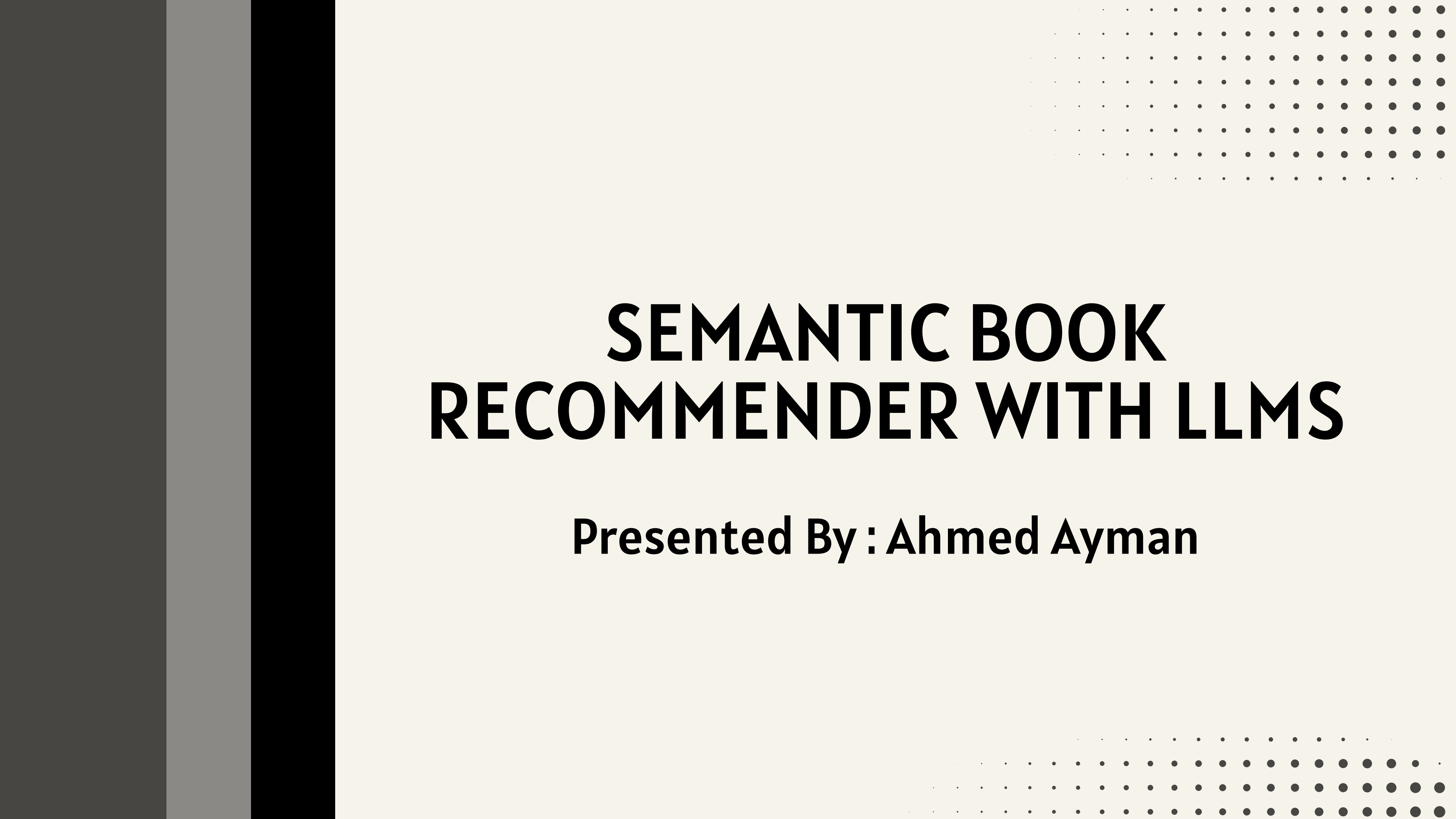
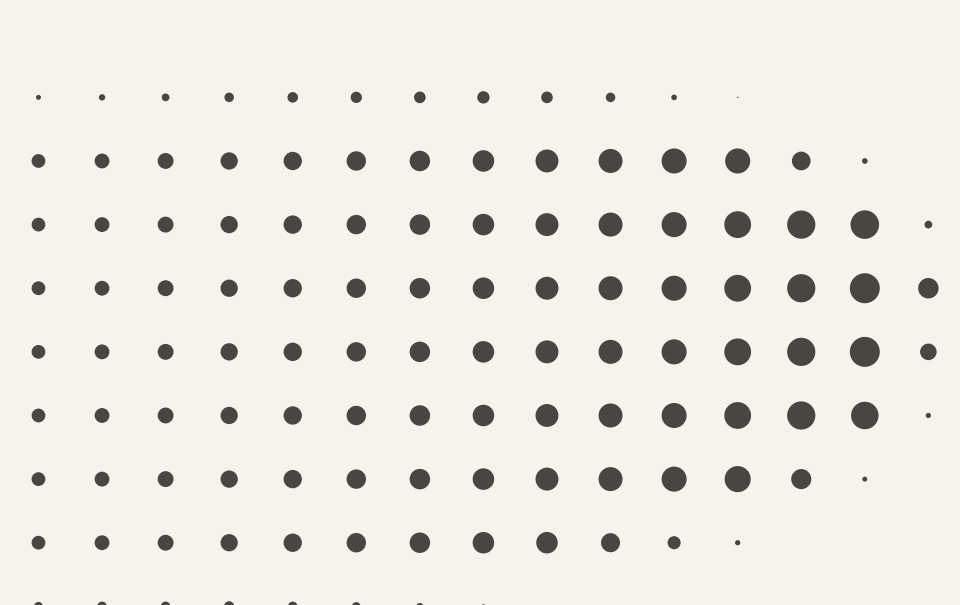


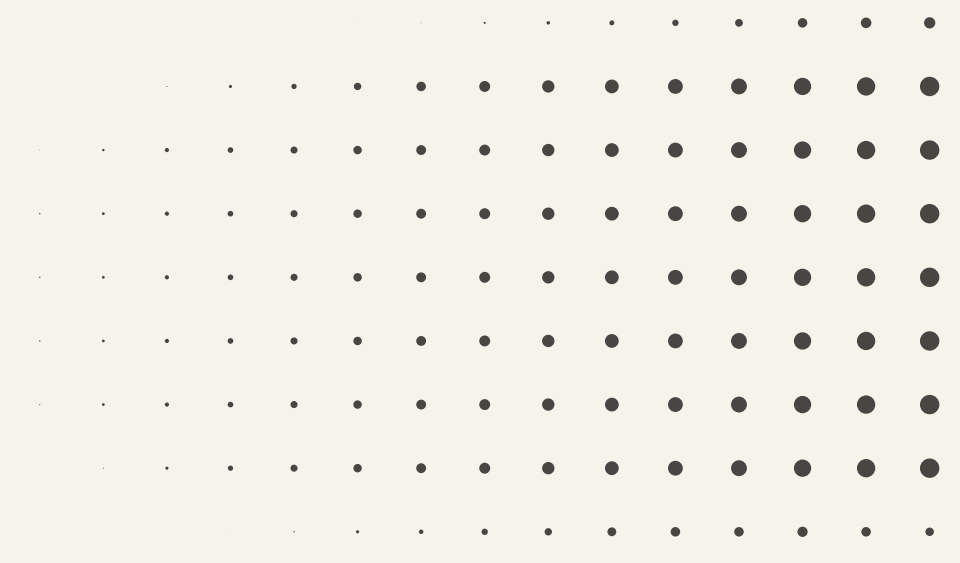
The slide features a decorative header with three vertical bars on the left (dark gray, medium gray, and black) and a grid of dots in the top right corner. The main title is centered in a large, bold, black font. Below the title, the presenter's name is displayed in a slightly smaller, bold, black font. The footer consists of a grid of dots in the bottom right corner.

SEMANTIC BOOK RECOMMENDER WITH LLMS

Presented By : Ahmed Ayman



OVERVIEW

- Overview
 - Project Objectives
 - Project Components (1/2)
 - Project Components (2/2)
 - Tools and Technologies Used
 - Challenges & Solutions
 - Outcomes
- 

OVERVIEW

This project implements a semantic book recommendation engine using LangChain, HuggingFace embeddings, and Chroma vector store. It enables natural language search where users find books based on description similarity rather than exact keywords. Results are refined through category filtering and sorted by specific emotional tones such as joy, sadness, or suspense. A Gradio-based web interface displays the final recommendations with book covers, authors, and truncated summaries.

PROJECT OBJECTIVES

1

Allows users to search for books using natural language queries (e.g., "a book about a person seeking revenge").

2

Classify books such as "fiction" or "non-fiction" for filtering.

3

Analyzes the sentiment and emotions of book descriptions to enable sorting by tone (e.g., suspenseful, joyful, sad).

4

Provides a user-friendly web interface for seamless interaction.

PROJECT COMPONENTS (1/2)

● Text Data Cleaning

- Dataset of 7,000 books from Kaggle.
- Handled missing values and added new features (e.g., book age).
- Visualized data and prepared it for analysis.

● Semantic (Vector) Search and Vector Database Creation

- Used sentence-transformers/all-mpnet-base-v2 to generate text embeddings.
- Stored vectors in ChromaDB for fast similarity search.
- Implemented top-10 book retrieval based on semantic similarity.

PROJECT COMPONENTS (2/2)

● Text Classification Using Zero-Shot Classification

- Used Facebook/BART-large-mnli model to classify books into fiction/non-fiction.
- Optimized for GPU/CPU performance.

● Sentiment Analysis Using LLMs

- Extracted emotions (joy, sadness, suspense) from book descriptions.
- Enabled users to sort books based on emotional tone.

● Web Application Development Using Gradio

- Designed a web interface integrating all functionalities.
- Deployed a working Gradio-based book recommender system.

01-DATA-EXPLORATION

- 1 Dataset Loading:** Downloaded the 7k Books with Metadata dataset from Kaggle and explored book features (ratings, pages, categories, descriptions).
- 2 EDA:** Analyzed statistics and visualized missing values; Fiction emerged as the most common category.
- 3 Feature Engineering:** Created new features like book age, description length, and missing-description indicator.
- 4 Data Cleaning:** Removed rows with missing key fields and filtered out very short descriptions; merged title + subtitle.
- 5 Final Output:** Performed correlation analysis and exported a cleaned dataset ready for NLP or recommendation models.

02-VECTOR-SEARCH

- 1 **RAG Pipeline Setup:** Installed LangChain tools and loaded the cleaned books dataset to prepare text for semantic search.
- 2 **Text Processing:** Converted descriptions into a text file, split them into chunks using recursive text splitting.
- 3 **Embeddings & Vector DB:** Generated embeddings with Hugging Face (all-MiniLM-L6-v2) and stored them in the → Chroma vector database.
- 4 **Semantic Search:** Used similarity search to retrieve books based on meaning (e.g., "books to teach children about nature").
- 5 **Recommendation Function:** Extracted ISBNs from results and returned matching books as semantic recommendations.

03-TEXT-CLASSIFICATION

- 1 **Category Simplification:** Mapped many detailed genres into broader groups (Fiction, Nonfiction, Children's categories).
- 2 **AI Classification:** Used zero-shot classification with the Hugging Face model → Facebook AI Research BART-large-MNLI to predict missing categories from book descriptions.
- 3 **Evaluation:** Tested predictions on sample data and achieved ~78% accuracy.
- 4 **Category Completion:** Filled missing labels using model predictions to create a fully categorized dataset.
- 5 **Final Output:** Saved the enriched dataset (books_with_categories.csv) for recommendation and NLP tasks.

04-SENTIMENT-ANALYSIS

- 1 **Emotion Detection:** Applied text-classification using the Hugging Face model → j-hartmann emotion-english-distilroberta-base to analyze book descriptions.
- 2 **Sentence-Level Analysis:** Split each description into sentences to capture fine-grained emotional signals.
- 3 **Score Aggregation:** Calculated max scores per emotion (joy, fear, sadness, etc.) for each book.
- 4 **Dataset Enrichment:** Merged emotion features with the original books dataset.
- 5 **Output:** Saved the final dataset (books_with_emotions.csv) for emotion-aware recommendations.

05-GRADIO-INTERFACE

- 1 **Semantic Search** – Finds books matching user queries using HuggingFace embeddings.
- 2 **Emotion Filtering** – Rank books by emotional tone: Happy, Sad, Angry, Surprising, or Suspenseful.
- 3 **Category Filter** – Narrow recommendations by book category (Fiction, Nonfiction, etc.).
- 4 **User-Friendly UI** – Gradio gallery with book thumbnails, authors, and truncated descriptions.
- 5 **Fast & Scalable** – Processes thousands of books efficiently with vector store for semantic search.

TOOLS AND TECHNOLOGIES USED

- **Programming Language:**
Python

- **Libraries & Frameworks:**
- Hugging Face Transformers
- Sentence-Transformers
- Pandas, NumPy
- ChromaDB, Langchain
- Gradio

- **Deployment:**
- Gradio Web App
- Space on Huggingface

CHALLENGES & SOLUTIONS

First Problem

Handling large text datasets: Used ChromaDB for efficient vector storage.

Second Problem

Classifying books without labeled data: Used Zero-Shot Classification with LLMs.

Third Problem

Building a user-friendly interface: Developed a Gradio web application.

OUTCOMES

- 1** Successfully built a working book recommender system.
- 2** Users can search books using natural language queries.
- 3** Books can be filtered by category (fiction/non-fiction).
- 4** Sentiment-based sorting allows users to discover books by emotional tone.
- 5** Fully functional web application deployed using Gradio.

FINAL OUTCOME

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Semantic book recommender

Please enter a description of a book:

e.g., A story about forgiveness

Select a category:

All

Select an emotional tone:

All

Find recommendations

Recommendations

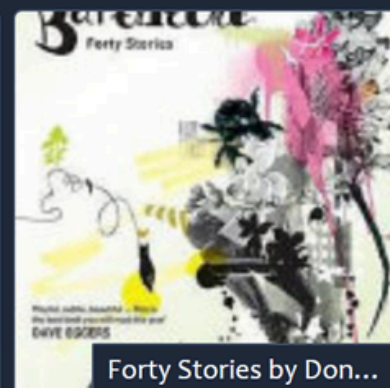
Recommended books



Good Omens by Neil...



How to Travel with ...



Forty Stories by Don...



"A" is for Abductive ...



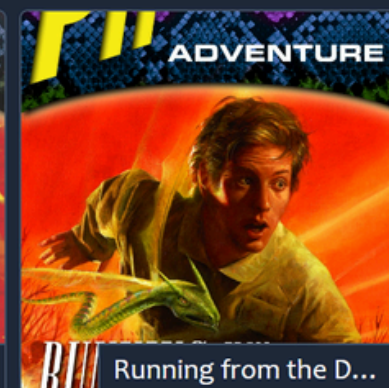
Lurulu by Jack Vanc...



Friday by Robert An...



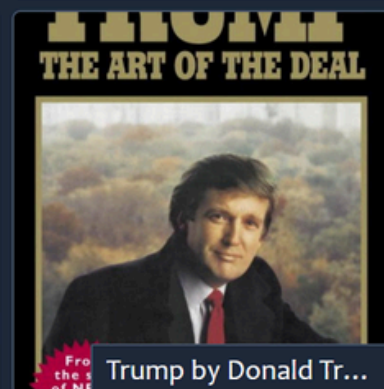
Flinx's Folly by Alan ...



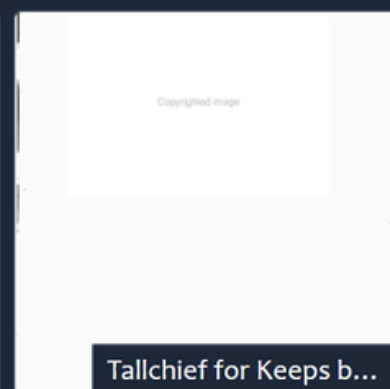
Running from the D...



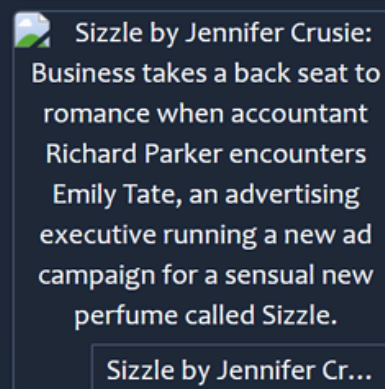
No Dominion by Cha...



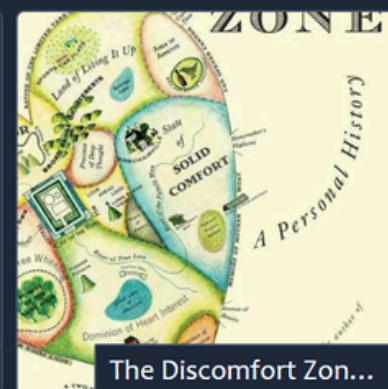
Trump by Donald Tr...



Tallchief for Keeps b...



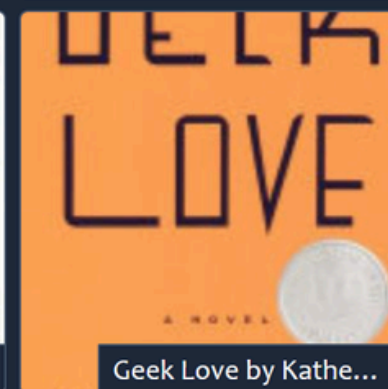
Sizzle by Jennifer Cr...



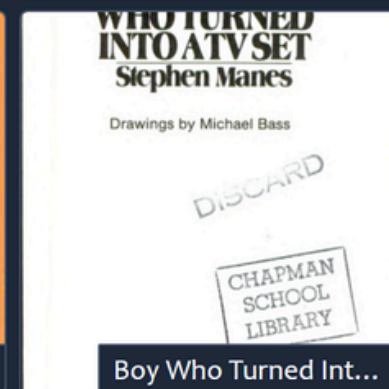
The Discomfort Zon...



Giving Good Weight ...



Geek Love by Kathe...



Boy Who Turned Int...

APP LINK → <https://huggingface.co/spaces/ahmed-ayman/book-recommender>

The image features a minimalist design with a light gray background. On the left side, there is a vertical bar composed of three segments: a dark gray segment at the top, a medium gray segment in the middle, and a light gray segment at the bottom. In the top right corner, there is a rectangular area filled with a grid of small black dots. A similar, but slightly larger and more dense, grid of black dots is located in the bottom right corner. Centered in the middle of the image is the text "THANK YOU" in a large, bold, black, sans-serif font.

THANK YOU